



SPATIAL INTELLIGENCE

Graph Analysis Methods for Architectural Analysis

Professor Wassim Jabi, PhD

INTRODUCTION

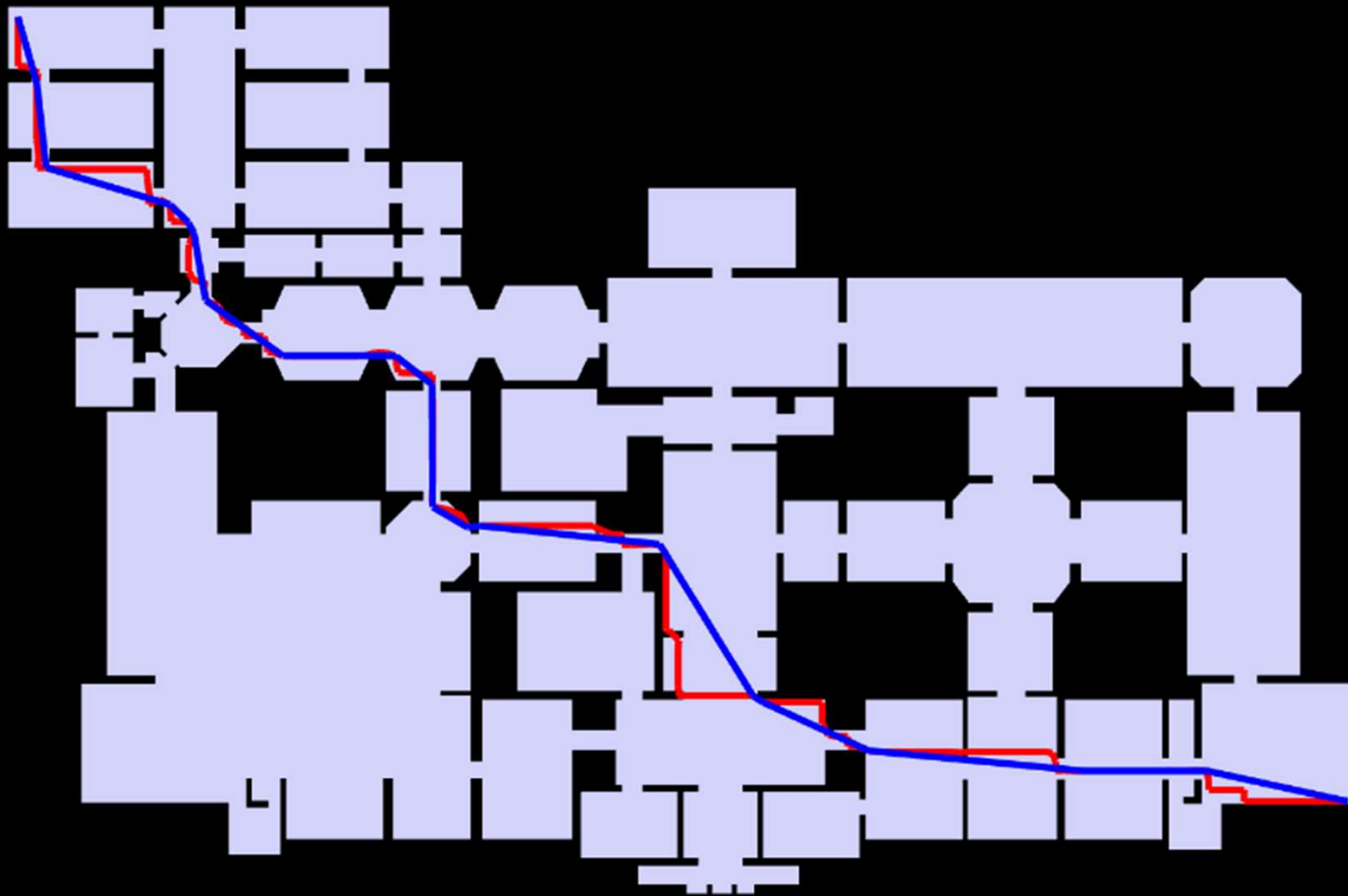
- Spatial intelligence in architecture refers to the ability to analyse and reason about spatial relationships using computational representations such as graphs.
- Buildings encode relational structures (adjacency, visibility, accessibility)
- Graphs provide a formal structure to analyse these relationships
- Enables transition from qualitative intuition → quantitative evaluation
- Core to:
 - Space syntax
 - Circulation analysis
 - Performance-driven design
- In TopologicPy: graphs act as a unifying analytical layer between geometry and semantics

OVERVIEW OF GRAPH ANALYSIS METHODS

- Shortest path
- Minimum spanning tree (MST)
- Graph density
- Graph diameter
- Degree Centrality
- Closeness centrality
- Betweenness centrality
- Visibility graph analysis
- Graph isomorphism
- Graph similarity

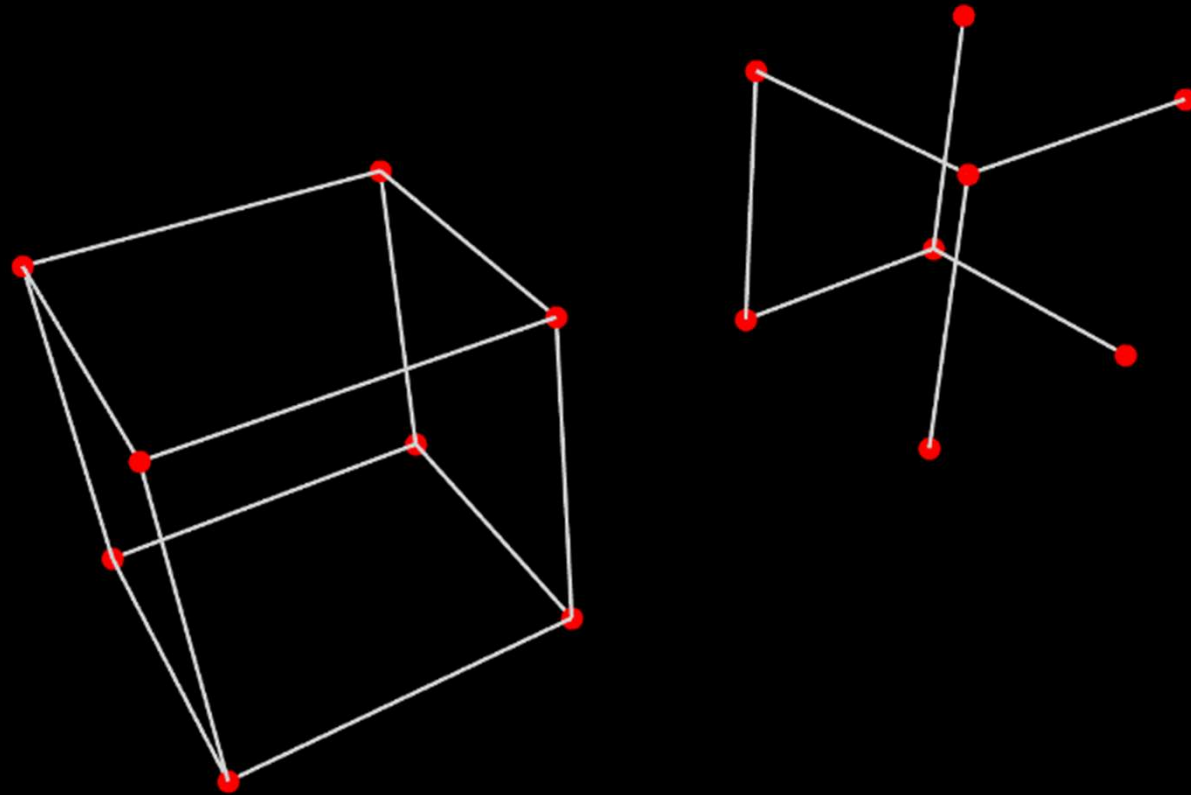
SHORTEST PATH

- Definition
 - The shortest path is the minimum-cost route between two vertices in a graph..
- Computation
 - Dijkstra
 - Breadth-First Search (BFS).
 - Cost can be defined by: Distance, Time, Angular change (space syntax), other Costs.
- Architectural Use:
 - Wayfinding analysis
 - Emergency evacuation
 - Accessibility studies



MINIMUM SPANNING TREE (MST)

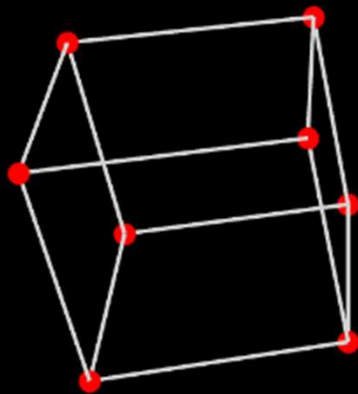
- Definition
 - A minimum spanning tree is a subset of edges that connects all vertices with minimum total edge weight and no cycles.
- Computation
 - Kruskal's algorithm
 - Prim's algorithm
- Architectural Use:
 - Identifies efficient backbone circulation networks
 - Useful for Infrastructure routing, Structural logic abstraction, Simplifying complex layouts



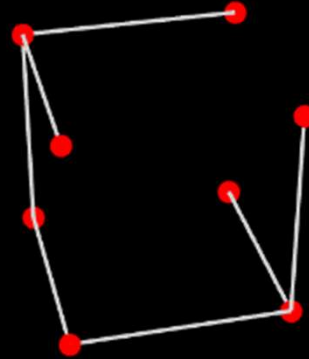
GRAPH DENSITY

- Definition
 - Graph density measures the proportion of actual connections relative to all possible connections in a graph.
- Computation
 - Density = $\frac{2E}{V(V-1)}$ for undirected graphs
 - Compare number of edges to complete graph
 - Values range from 0 (sparse) to 1 (fully connected)
- Architectural Use:
 - Indicates degree of spatial interconnectedness
 - High density → highly permeable layouts
 - Low density → segregated or hierarchical layouts

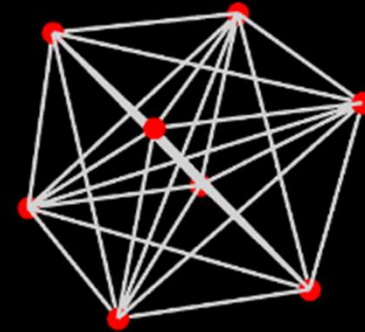
GRAPH DENSITY



0.43



0.25

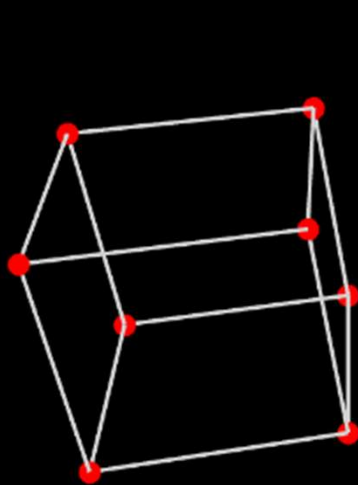


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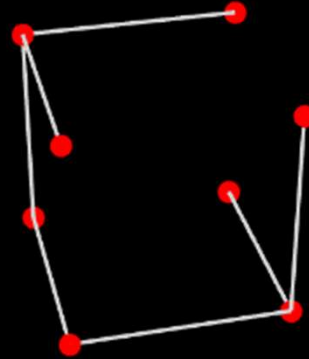
GRAPH DIAMETER

- Definition
 - Graph diameter is the longest shortest path between any pair of vertices in a graph.
- Computation
 - Compute all-pairs shortest paths
 - Identify the maximum shortest path distance
- Architectural Use:
 - Measures overall spatial depth
 - Large diameter → deep, segregated buildings
 - Small diameter → compact, accessible layouts

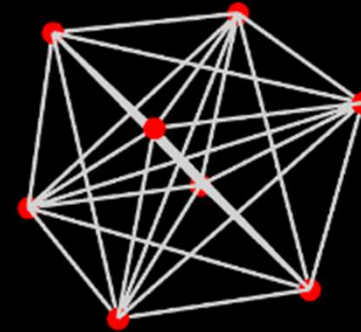
GRAPH DIAMETER



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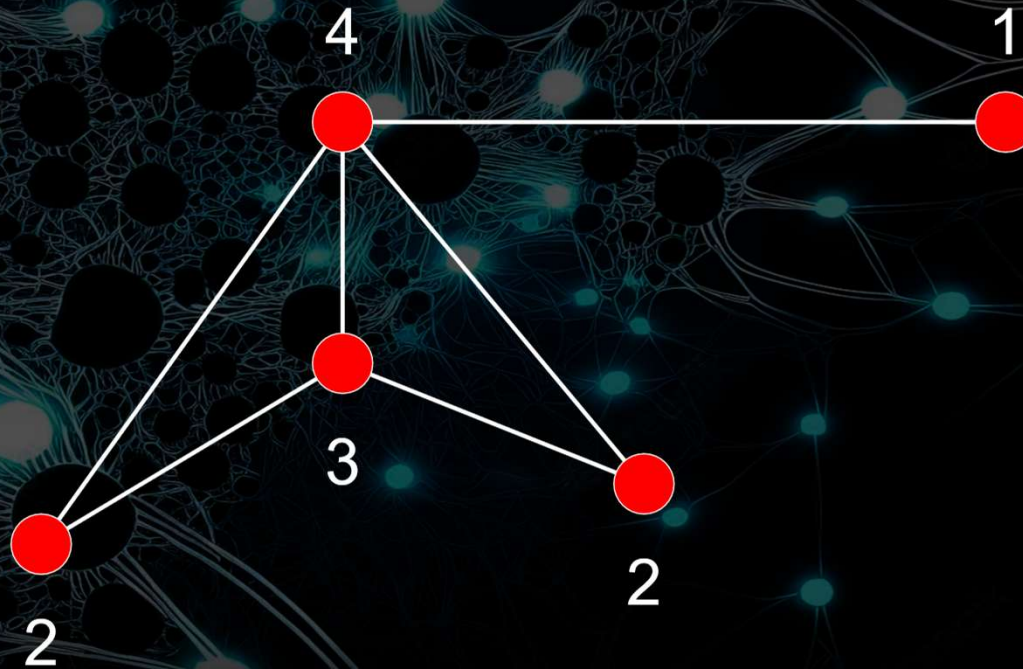


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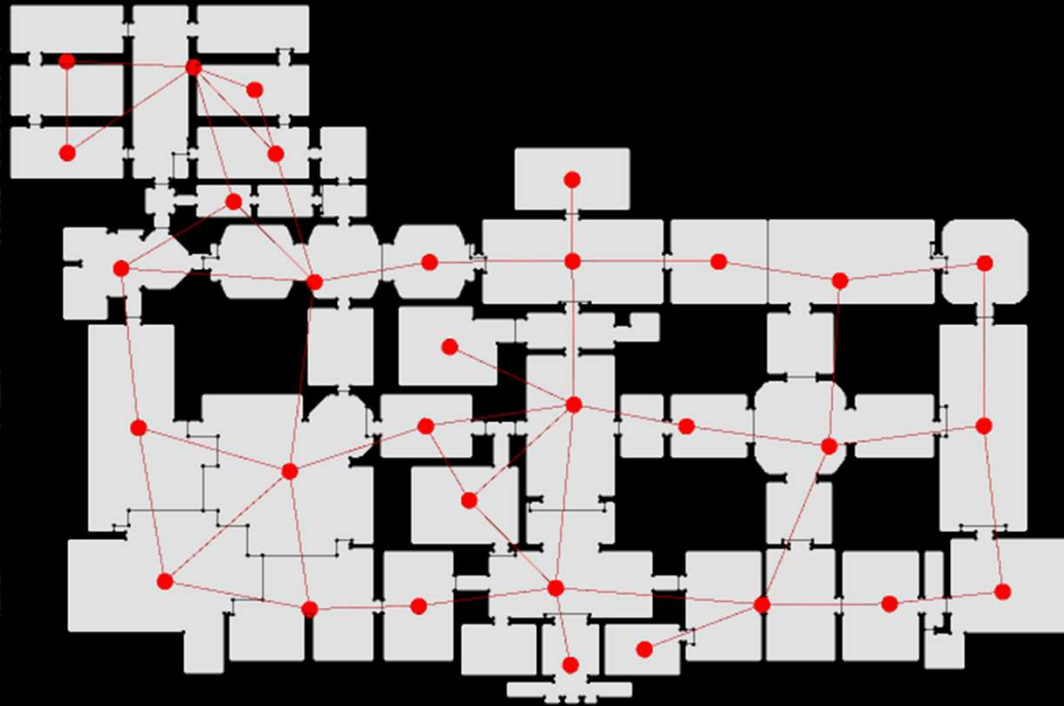
DEGREE CENTRALITY (CONNECTIVITY)

- Definition
 - Degree centrality measures the number of direct connections (edges) a node has to other nodes in the graph.
- Computation
 - For each node, count the number of incident edges
 - For weighted graphs, optionally sum edge weights instead of counting
 - Can be normalised by dividing by $V-1$.
 - In spatial graphs: Often corresponds to immediate neighbours (adjacent or visible spaces)
- Architectural Use:
 - Measures local connectivity / permeability
 - High degree → highly connected, accessible spaces
 - Low degree → isolated or controlled spaces
 - In space syntax: Known as connectivity
 - Strong predictor of movement intensity at local scale

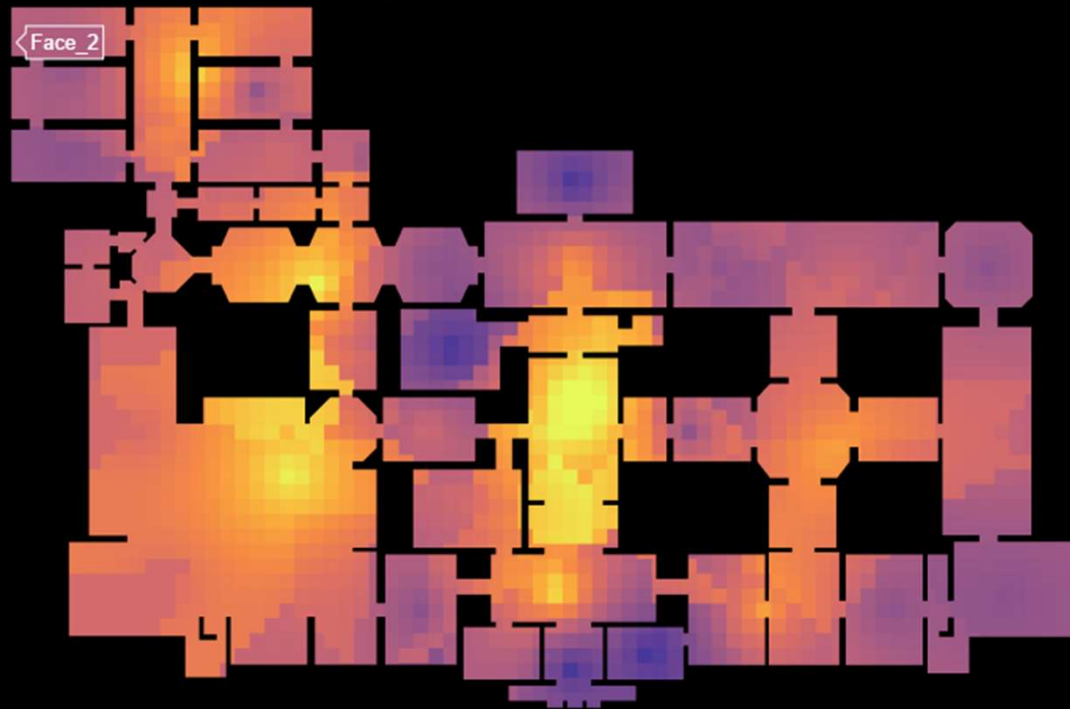
DEGREE CENTRALITY (CONNECTIVITY)



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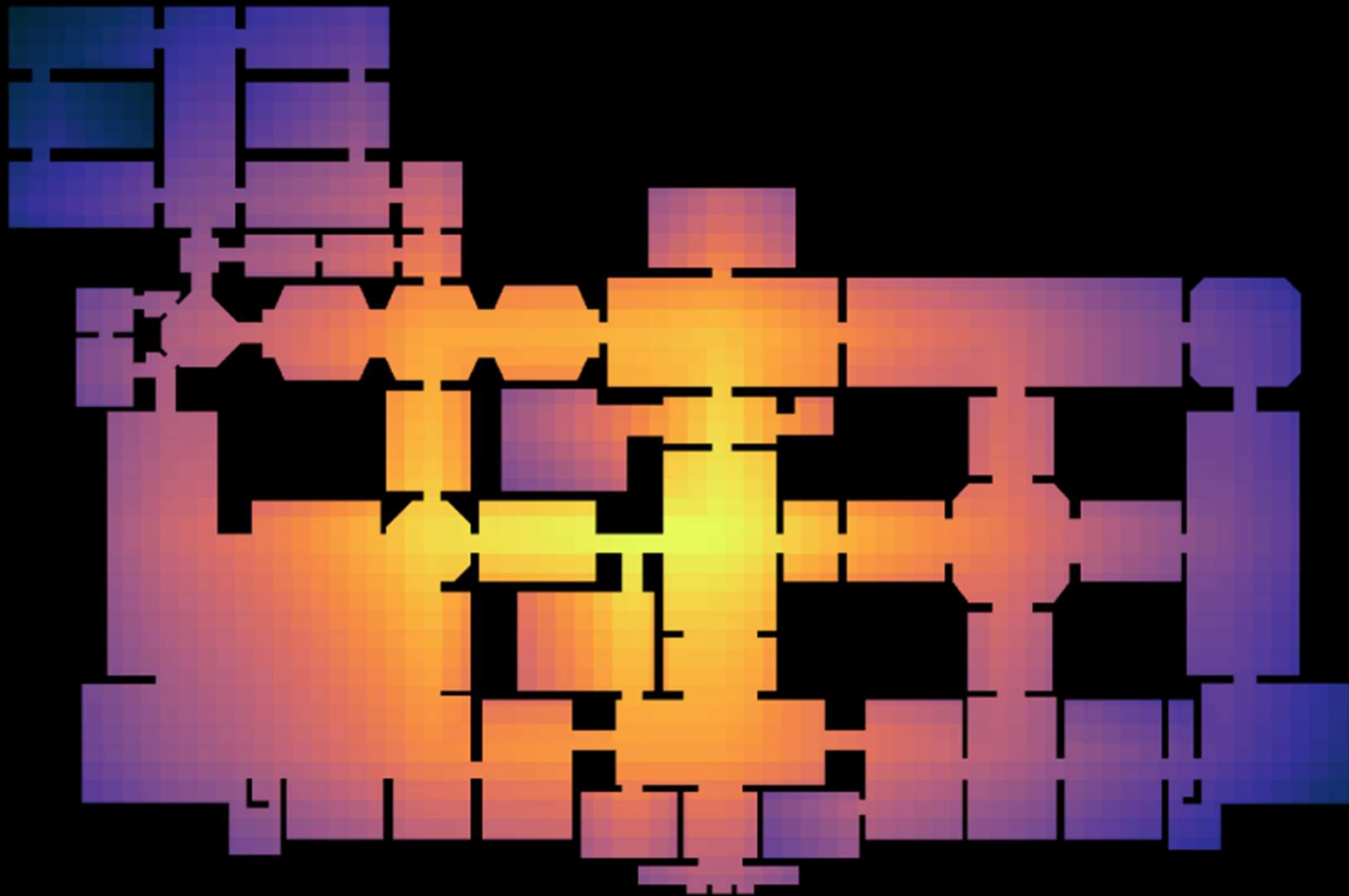


DEGREE CENTRALITY (CONNECTIVITY)



CLOSENESS CENTRALITY

- Definition
 - Closeness centrality measures how close a node is to all other nodes in the graph.
- Computation
 - Inverse of sum of shortest path distances to all other nodes
 - $$C(v) = \frac{1}{\sum d(v,u)}$$
- Architectural Use:
 - Identifies spatial accessibility
 - High closeness → central, easily reachable spaces
 - Often corresponds to: Lobbies, Hubs, Circulation cores



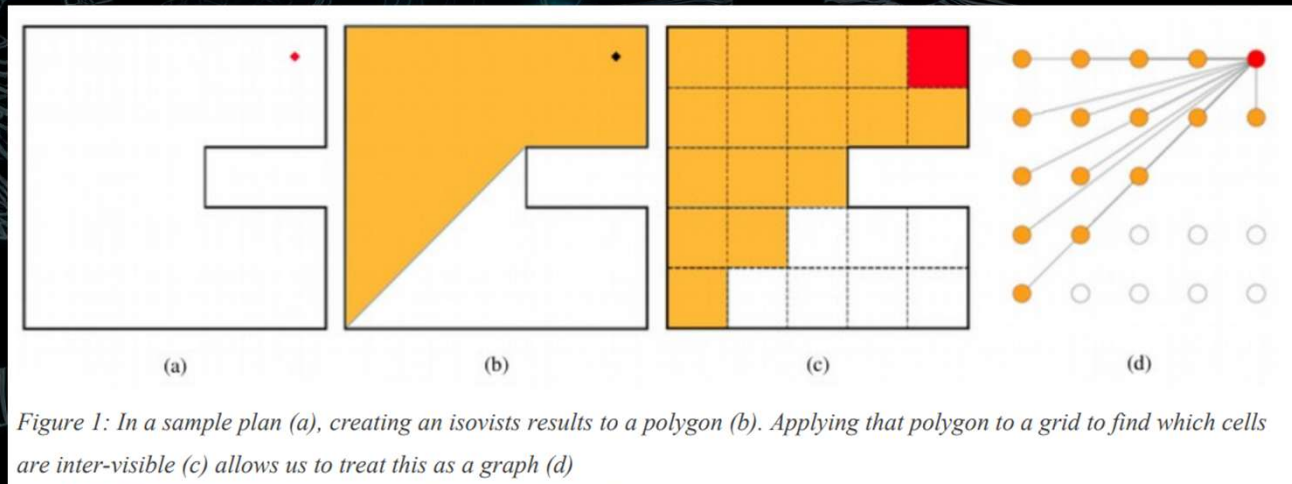
BETWEENNESS CENTRALITY

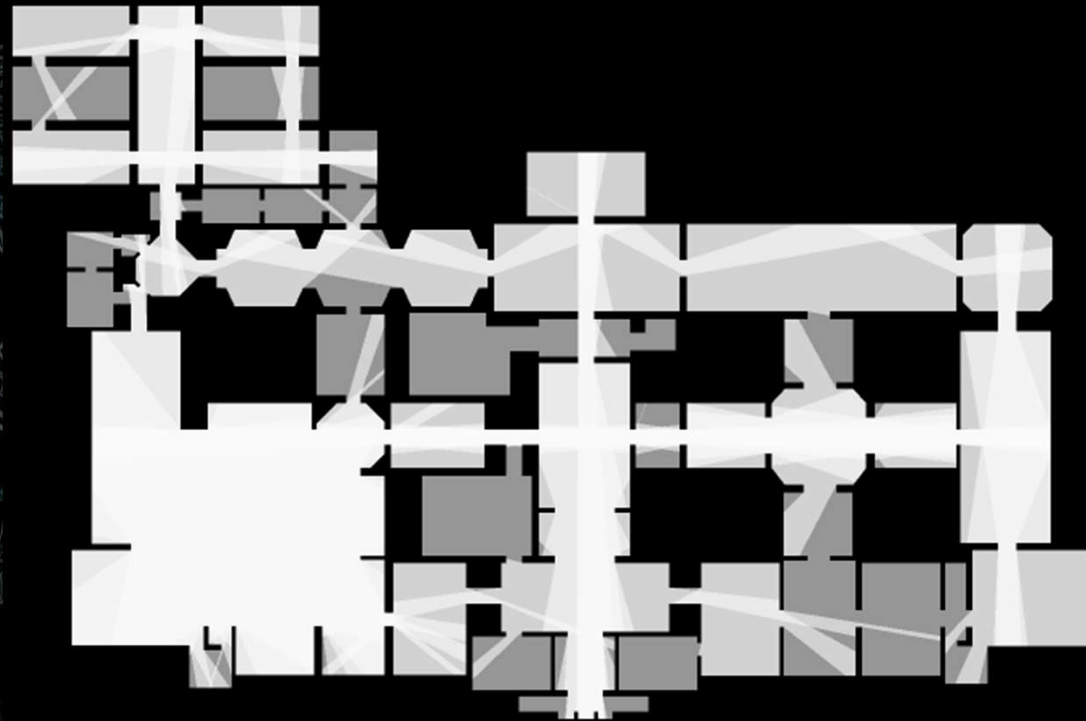
- Definition
 - Betweenness centrality measures how often a node lies on the shortest paths between other nodes.
- Computation
 - Count number of shortest paths passing through each node
 - Typically computed using Brandes algorithm
- Architectural Use:
 - Identifies movement bottlenecks and critical connectors
 - High betweenness → corridors, thresholds, gateways
 - Useful for: flow optimisation, congestion analysis

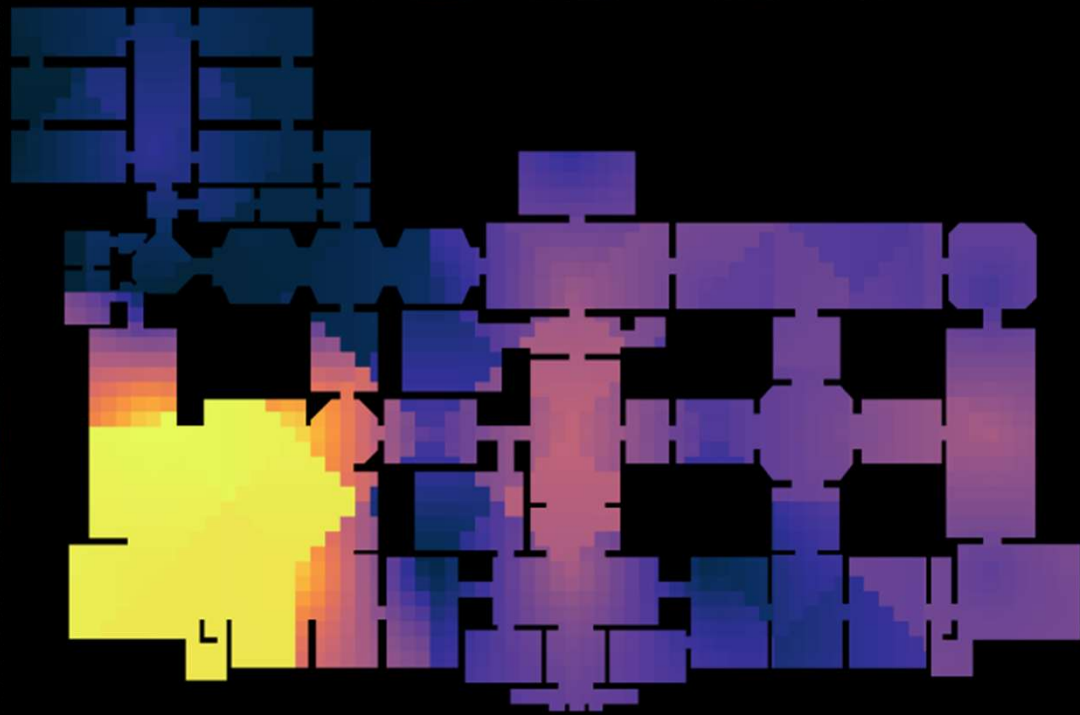


VISIBILITY GRAPH ANALYSIS (VGA)

- Definition
 - Visibility Graph Analysis represents space as a graph where nodes are sample points and edges connect points that are mutually visible.
- Computation
 - Discretise space into a grid of sample points (typically uniform)
 - For each point:
 - Compute line-of-sight to all other points
 - Create edges between mutually visible points
 - Construct a visibility graph
 - Apply graph metrics (e.g. connectivity, clustering, centrality)
- Architectural Use:
 - Quantifies visual accessibility and spatial perception
 - Used to evaluate: wayfinding clarity, surveillance potential, spatial openness vs enclosure
 - Forms the computational basis of space syntax VGA studies (e.g. in DepthmapX)







GRAPH ISOMORPHISM

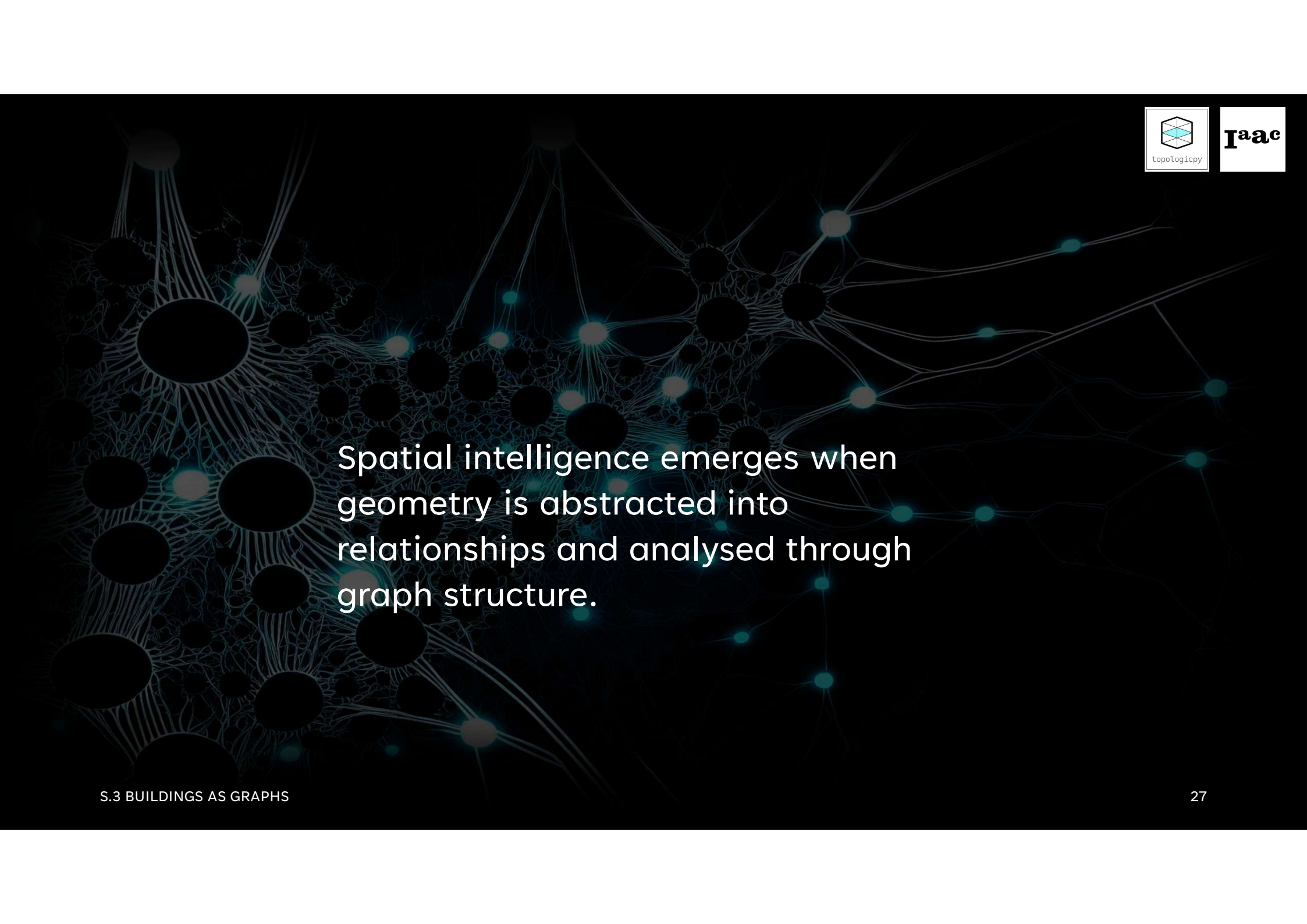
- Definition
 - Graph isomorphism determines whether two graphs have identical structure regardless of node labels.
- Computation
 - Compare adjacency structures
 - Use canonical labelling or backtracking algorithms
- Architectural Use:
 - Identifies topologically equivalent layouts
 - Useful for: precedent comparison, pattern recognition, typological studies

GRAPH SIMILARITY

- Definition
 - Graph similarity measures how structurally similar two graphs are, even if they are not identical.
- Computation
 - Methods:
 - Graph edit distances
 - Spectral similarity
 - Embedding-based similarity
 - Compare structure, attributes, or subgraphs
- Architectural Use:
 - Enables: precedent retrieval, case-based design, dataset comparison (e.g., housing plans)
 - Critical for GraphRAG-style workflows

SYNTHESIS

- Graph analysis transforms architecture into a computable system
- Each metric reveals a different dimension of spatial intelligence:
 - Accessibility
 - Hierarchy
 - Visibility
 - Clustering / Classification
- Together, they form a multi-layer analytical framework



Spatial intelligence emerges when geometry is abstracted into relationships and analysed through graph structure.

THANK YOU

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